

SILICON PHOTONICS DESIGN, FABRICATION AND DATA ANALYSIS

Design and Simulation of Silicon Photonic Building Blocks on an SOI Platform

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ECE. 26'

Abstract

This report presents the design, simulation, and layout implementation of several fundamental silicon photonic integrated circuit (PIC) building blocks using the Lumerical MODE Solutions, INTERCONNECT, and KLayout/SiEPIC workflow. The work focused on waveguide modal analysis, polarization characterization, passive optical splitter design, Mach–Zehnder interferometer (MZI) implementation, Michelson interferometer analysis, and photonic mask layout generation on a silicon-on-insulator (SOI) platform.

The first section investigates guided modes in silicon strip waveguides using finite-difference eigenmode (FDE) analysis. Effective indices, polarization fractions, and field confinement characteristics for quasi-TE and quasi-TM modes were studied. A MATLAB slab-waveguide solver was additionally used to analytically estimate supported modes and compare theoretical results with numerical simulations.

The second section explores passive photonic circuit design in INTERCONNECT, including Y-branch splitters, 20:80 optical splitters, Mach–Zehnder interferometers, and Michelson interferometers. Transmission responses, interference behavior, and optical power distributions were analyzed using compact models and S-parameter simulations.

Finally, photonic mask layouts were generated using the SiEPIC EBeam process design kit (PDK) in KLayout. Multiple MZI structures, calibration waveguides, and Bragg grating-inspired structures were designed hierarchically and prepared for fabrication-oriented layout verification.

Index Terms—Silicon photonics, SOI waveguides, FDE analysis, Mach–Zehnder interferometer, Michelson interferometer, Bragg gratings, INTERCONNECT, KLayout, SiEPIC.

I. INTRODUCTION

Silicon photonics has emerged as one of the leading platforms for integrated optical systems due to its compatibility with mature CMOS fabrication infrastructure, high refractive-index contrast, and scalability for mass manufacturing. By integrating optical components onto compact silicon chips, silicon photonics enables high-bandwidth communication systems, optical sensing, quantum photonics, and photonic computing applications.

One of the primary advantages of silicon photonics is the strong optical confinement provided by the silicon-on-insulator (SOI) platform. Silicon possesses a refractive index of approximately 3.47 near 1550 nm, while silicon dioxide (SiO₂) has an index near 1.44. This large refractive-index contrast enables submicron optical confinement and compact photonic device geometries.

The work performed in this project focused on understanding and implementing several foundational photonic devices commonly used in integrated optical systems:

- Silicon strip waveguides
- Y-branch splitters
- 20:80 optical splitters
- Mach–Zehnder interferometers (MZIs)
- Michelson interferometers
- Calibration structures
- Bragg grating-inspired periodic structures

The overall design workflow consisted of:

1. Performing modal analysis using Lumerical MODE Solutions.
 2. Extracting effective indices and polarization properties.
 3. Building compact photonic circuits using INTERCONNECT.
 4. Designing fabrication-oriented layouts using KLayout and the SiEPIC EBeam PDK.
 5. Exporting fabrication-ready GDS layouts.
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II. THEORY

A. Silicon Strip Waveguides

The waveguide geometry considered throughout this work is the standard SOI strip waveguide consisting of:

- 220 nm silicon device layer

- 2 μm buried oxide (BOX)
- Silicon dioxide upper cladding
- 500 nm waveguide width

The large refractive-index contrast between silicon and silicon dioxide enables strong confinement of guided optical modes.

For guided propagation to occur, total internal reflection must confine the optical field inside the silicon core. The propagation constant is given by

$$[\beta = k_0 n_{\text{eff}}]$$

where:

- β is the propagation constant
- $k_0 = 2\pi/\lambda$
- n_{eff} is the effective refractive index

The effective index lies between the cladding and core refractive indices:

$$[n_{\text{cladding}} < n_{\text{eff}} < n_{\text{core}}]$$

B. Quasi-TE and Quasi-TM Polarizations

Due to the rectangular geometry of silicon strip waveguides, the supported optical modes are not perfectly transverse electric (TE) or transverse magnetic (TM). Instead, they are referred to as quasi-TE and quasi-TM modes.

For quasi-TE modes:

- Dominant electric-field component: (E_y)
- Major field polarization lies parallel to the wafer surface

For quasi-TM modes:

- Dominant magnetic-field component: (H_y)
- Dominant electric field is oriented vertically

The polarization characteristics were examined using Lumerical MODE Solutions and compared against instructional material from the silicon photonics course notes.

C. Mach–Zehnder Interferometer

The Mach–Zehnder interferometer (MZI) is one of the most important structures in integrated photonics.

A basic MZI consists of:

1. Input splitter
2. Two optical arms
3. Output combiner

An optical phase difference between the two arms produces interference at the output.

The optical intensity response is given by:

$$I_{\text{out}} = I_{\text{in}}^2()$$

where the phase difference is

$$\Delta\phi = kL$$

with:

- k : propagation constant
- L : path-length difference

Constructive interference occurs when:

$$\Delta\phi = 2m\pi$$

and destructive interference occurs when:

$$\Delta\phi = (2m+1)\pi$$

MZIs are widely used in:

- Optical modulators
- Wavelength filters
- Optical switches
- Sensors
- Delay-line structures

D. Michelson Interferometer

The Michelson interferometer uses reflective optical paths to generate interference.

Unlike the MZI, where light propagates through separate transmission paths, the Michelson interferometer reflects light back toward the splitter after propagating through unequal arm lengths.

The resulting interference pattern depends on the relative optical phase accumulated during the round-trip propagation.

Michelson interferometers are commonly used in:

- Precision metrology
 - Optical coherence systems
 - Interferometric sensing
 - Spectroscopy
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E. Optical Splitters

Optical splitters divide optical power between multiple outputs.

A 50:50 splitter ideally divides optical power equally, while asymmetric splitters such as 20:80 structures intentionally distribute power unevenly.

The power splitting ratio is defined as:

[=]

where:

- (P_{out}): output optical power
- (P_{in}): input optical power

Y-branch splitters and directional couplers are commonly used implementations.

III. WAVEGUIDE MODELING AND FDE ANALYSIS

A. FDE Simulation Setup

Waveguide modal analysis was performed using Lumerical MODE Solutions with the finite-difference eigenmode (FDE) solver.

Simulation parameters included:

Parameter	Value
Wavelength	1550 nm
Silicon thickness	220 nm
Waveguide width	500 nm
Core index	3.47
Cladding index	1.444

The simulation investigated:

- Effective index
 - Mode confinement
 - Polarization fractions
 - Supported modes
 - Field distributions
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B. Quasi-TE Mode Analysis

The fundamental quasi-TE mode demonstrated strong optical confinement within the silicon core region.

The dominant electric-field component was observed along the horizontal direction parallel to the wafer surface. The mode exhibited a high TE polarization fraction consistent with theoretical expectations for a strip waveguide geometry.

Simulation results showed:

Mode	Effective Index
TE1	~2.44

The optical field decayed evanescently into the surrounding oxide cladding.

C. Quasi-TM Mode Analysis

The quasi-TM mode exhibited weaker confinement compared to the quasi-TE mode due to the lower vertical confinement of the electric field.

Simulation results showed:

Mode	Effective Index
TM1	~1.78

The TM mode displayed increased field penetration into the oxide cladding.

D. MATLAB Slab-Waveguide Analysis

A MATLAB slab-waveguide solver was additionally studied and implemented to analytically compute waveguide modes and effective indices.

The MATLAB implementation solved the slab-waveguide transcendental equations for both TE and TM polarizations and plotted the corresponding field distributions.

The script computed:

- Effective refractive indices
- TE and TM mode profiles
- Propagation constants
- Field confinement behavior

The analytical MATLAB results agreed closely with the numerical FDE simulations performed in Lumerical MODE.

IV. INTERCONNECT SIMULATIONS

A. Y-Branch Splitter

A Y-branch optical splitter was constructed in INTERCONNECT using compact photonic models.

The device divided optical power between two output waveguides. Ideally, the splitter provides equal power distribution between both outputs while minimizing insertion loss.

The optical transmission characteristics were analyzed using wavelength sweeps across the C-band region.

The simulated splitter demonstrated:

- Stable transmission response
 - Low insertion loss
 - Proper optical splitting behavior
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B. 20:80 Optical Splitter

An asymmetric optical splitter was designed to distribute optical power unequally between the output ports.

The target design ratio was:

- 20% optical power in one output arm
- 80% optical power in the second output arm

This structure demonstrated controlled optical power division and highlighted the importance of coupling-region geometry in determining splitter ratios.

The simulated transmission results confirmed the asymmetric power distribution.

C. Mach–Zehnder Interferometer

A Mach–Zehnder interferometer was implemented using:

- Input splitter
- Two optical arms
- Output combiner

A path-length imbalance was introduced between the two arms to generate wavelength-dependent interference.

The transmission spectrum displayed periodic constructive and destructive interference behavior consistent with the theoretical MZI transfer function.

The observed interference fringes demonstrated:

- Optical phase sensitivity
- Path-length dependence
- Wavelength filtering characteristics

The MZI structure forms one of the most fundamental integrated photonic building blocks.

D. Michelson Interferometer

A Michelson interferometer was additionally modeled in INTERCONNECT.

The structure reflected optical signals back toward the splitter after propagating through different optical path lengths.

Interference patterns resulting from the reflected optical fields were analyzed using transmission monitors.

The simulation demonstrated:

- Interference sensitivity to path imbalance
 - Reflection-based interference operation
 - Optical coherence behavior
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V. KLAYOUT AND PHOTONIC MASK DESIGN

A. SiEPIC EBeam Workflow

Photonic mask layouts were generated using:

- KLayout
- SiEPIC tools
- EBeam process design kit (PDK)

The workflow enabled:

- Hierarchical PIC design
- Parametric cell generation
- Optical routing
- Functional verification
- GDS export

Several layout structures were designed including:

- Multiple MZIs
 - Calibration waveguides
 - Bragg grating-inspired devices
 - Optical splitter calibration structures
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B. Mach–Zehnder Layouts

Multiple MZI structures were designed with varying optical path lengths.

The layouts incorporated:

- Grating couplers
- Waveguide bends
- Delay-line sections
- Routing structures

The layouts were verified using the SiEPIC functional verification tools.

C. Bragg Grating-Inspired Structures

Periodic waveguide perturbations were explored to create Bragg grating-inspired photonic structures.

Bragg gratings selectively reflect optical wavelengths satisfying the Bragg condition:

$$[B = 2 n_{\text{eff}} \Lambda]$$

where:

- Λ : Bragg wavelength
- n_{eff} : effective index
- Λ : grating period

The structures consisted of periodic corrugations along the waveguide path intended to demonstrate distributed optical reflection behavior.

VI. RESULTS AND DISCUSSION

The simulations and layouts developed throughout this work demonstrated the complete workflow for integrated silicon photonic circuit design.

The FDE simulations successfully identified:

- Quasi-TE and quasi-TM modes
- Effective refractive indices
- Optical confinement properties
- Polarization fractions

The INTERCONNECT simulations validated the operation of several important passive photonic devices including:

- Y-branch splitters
- 20:80 splitters
- Mach–Zehnder interferometers
- Michelson interferometers

The generated KLayout designs demonstrated successful implementation of hierarchical PIC layouts using the SiEPIC EBeam PDK workflow.

The project additionally provided practical understanding of:

- Photonic design-rule considerations
- Hierarchical layout structures
- Compact-model simulation workflows
- Optical interference phenomena
- Integrated photonic circuit routing

VII. CONCLUSION

This project explored the design, simulation, and layout implementation of several foundational silicon photonic devices using the Lumerical MODE, INTERCONNECT, and KLayout/SiEPIC workflow.

Waveguide modal analysis demonstrated the propagation characteristics of quasi-TE and quasi-TM modes in a silicon strip waveguide geometry. Effective indices and polarization behavior were successfully extracted using finite-difference eigenmode simulations and compared against analytical slab-waveguide calculations.

Passive photonic components including Y-branch splitters, 20:80 optical splitters, Mach-Zehnder interferometers, and Michelson interferometers were implemented and analyzed in INTERCONNECT.

Finally, fabrication-oriented layouts were generated using KLayout and the SiEPIC EBeam PDK. The resulting structures demonstrated practical integrated photonic design workflows and hierarchical layout techniques.

Overall, the project provided comprehensive exposure to modern silicon photonics design methodologies and integrated optical device simulation.

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APPENDIX A — SOFTWARE TOOLS USED

Software	Purpose
Lumerical MODE Solutions	FDE modal analysis
Lumerical INTERCONNECT	Photonic circuit simulation
KLayout	PIC mask layout design
SiEPIC Tools	PDK integration and verification
MATLAB	Analytical slab-waveguide calculations

APPENDIX B — FILES GENERATED DURING THE PROJECT

File	Description
EBeam_joyjustin.gds	Photonic mask layout
MZI.icp	Mach–Zehnder interferometer simulation
michson.icp	Michelson interferometer simulation
splits2080.icp	20:80 splitter simulation
wg1export.ldf	Waveguide FDE simulation export
wg_1D_slab.m	MATLAB slab-waveguide solver

NOTES

This report was prepared using simulation results and layouts generated through the silicon photonics tutorial workflow involving:

- Finite-difference eigenmode simulations
- Waveguide modal analysis
- INTERCONNECT photonic circuits
- SiEPIC EBeam process design kit layouts
- KLayout hierarchical photonic design